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Paul Haskins

Personal Statement

I am currently completing my undergraduate degree in Theoretical Physics at Trinity College Dublin, where I have developed a deep and enduring interest in the interplay between geometry, arithmetic, and high-energy theoretical physics. I hope to pursue a PhD under the supervision of Prof. Albrecht Klemm, whose work on quantum geometry, mirror symmetry, and the arithmetic of Calabi–Yau varieties aligns strongly with both my present academic direction and my broader intellectual motivations.

My undergraduate thesis, supervised by Prof. Jan Manschot concerns quantum black holes in the context of 2-dimensional conformal field theory. This work has given me exposure to the appearance of modular forms in physical settings, including partition functions, modular invariants, and the analytic structure of conformal blocks. During this research I encountered number-theoretic ideas through sources in various contexts such as [?, ?, ?]. These works helped reveal to me the remarkable unity between mathematical structures and physical phenomena, a theme central to Prof. Klemm’s contributions to the field.

I am particularly drawn to research that explores deep connections between enumerative geometry, topological string theory, and arithmetic structures. The role of modularity in counting BPS states, the arithmetic properties of periods, and the geometry of Calabi–Yau moduli spaces are areas I am especially excited to explore. I greatly admire Prof. Klemm’s influential work on topological string amplitudes, Gromov–Witten invariants, and the arithmetic of mirror symmetry, and I would be thrilled at the opportunity to contribute to this expanding framework in the context of the Leverhulme Trust project Quantum geometry and arithmetics. While my academic performance has at times reflected the challenges of balancing broad intellectual curiosity with formal coursework demands, I believe my progress in research demonstrates persistence, conceptual maturity, and genuine enthusiasm for theoretical physics. I thrive in environments of independent investigation and mathematical discussion, and I am eager to further develop the mathematical sophistication and technical depth required for work at the intersection of geometry and quantum theory. I would be honoured to join Prof. Klemm’s group and continue exploring these rich structures at the frontier of mathematical physics. Thank you for considering my application.